# Intruder Hunt

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## Overview

An intrusion detection system (IDS) is a device or software application that monitors a network or systems for malicious activity or policy violations. Any intrusion activity or violation is typically reported either to an administrator or collected centrally using a security information and event management (SIEM) system.

## 1 Lab Prep

Pull the lab related configuration files from the GitHub repository.

```
# Make sure that you are in the Docker_Security_Labs folder
# Use the pwd command can verify your current folder
git clone https://github.com/liranmatcu/Docker_Security_Labs.git
# or ``git pull'' if you have cloned the repository previously
```

Go to the lab folder under Docker_Security_Labs as follows,

```
cd  Docker_Security_Labs/ids/
```

build the Snort container image for this lab using docker-compose,

```
docker-compose build snort
```

and start Snort container instance as follows.

```
docker-compose run --rm snort bash
```

Testing configuration

```
snort -T -i eth0 -c /etc/snort/snort.conf
```

If everything goes smoothly, you should see Snort successfully validated the configuration!.

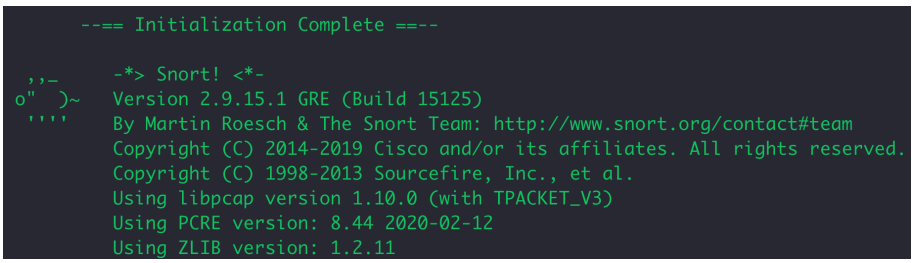


Figure 1: Snort initialization and test.

Open another terminal, start a container instance as a penetration tester.

```
docker-compose run --rm pentester bash
```

Then, we run a namp scan on the Snort container from the penetration tester container.

```
nmap snort
```

# 2 Snort Scenarios

## 2.1 Ping Flood Monitoring

Ping is a computer network administration software utility used to test the reachability of a host on an Internet Protocol (IP) network. It can be used by a simple denial-of-service attack where the attacker floods the victim with ICMP ”echo request” (ping) packets. The attacker hopes that the victim will respond with ICMP ”echo reply” packets, thus consuming both outgoing bandwidth as well as incoming bandwidth. If the target system is slow enough, it is possible to consume enough of its CPU cycles for a user to notice a significant slowdown.

We have added an rule in Snort configuration to detect ping scans. On your Snort container, issue the following command to start monitoring ping scans.

```
snort -i eth0 -c /etc/snort/snort.conf -A console
```

On the penetration tester container, start ping the Snort container as follows.

```
ping snort
```

The Snort container should detect the ping scan and display an alert message. An example screenshot is given Fig. 2.

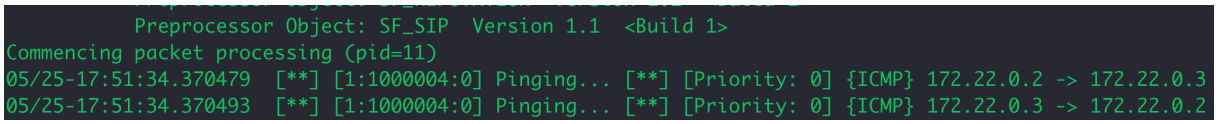


Figure 2: Ping alert.

### 2.1.1 Examine the local.rules file under the "./ids/snort/rules" folder on your local machine (it would appear under the /opt/snort/rules folder on the Snort container). Check the rule for ping flood detection. Can you improve the rule by adding more specifics?

## 2.2 SSH Brute Force Monitoring

On the penetration tester container, start an SSH attempt on the Snort container as follows.

```
nmap snort -p 22
```

The Snort container should detect the nmap SSH scan and display an alert message. An example screenshot is given Fig. 3.

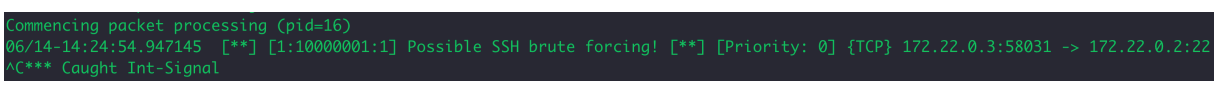


Figure 3: Ping alert.

### 2.2.1 Examine the local.rules file under the "./ids/snort/rules" folder on your local machine (it would appear under the /opt/snort/rules folder on the Snort container). Check the rule for SSH brute force detection. Can you improve the rule by adding more specifics?

### 2.2.2 Explain the basic syntax of Snort rules.

### 2.2.3 Examine the "./ids/snort/rules" folder on your local machine. Explain how to add new rules.

Reference: https://paginas.fe.up.pt/~mgi98020/pgr/writing_snort_rules.htm and https://snort.org/downloads#rules.

#### 2.2.4 Add the following rules and verify their effectiveness (you will need to provide screenshots).

- Detect nmap TCP pings. A nmap TCP ping sets this field to zero and sends a packet with the TCP ACK flag set to determine if a network host is active.
- Detect a SYN-FIN scan, where the reserved bits can be used to detect unusual behavior, such as IP stack fingerprinting attempts or other suspicious activity.
- Detect RPC requests and automatically decodes the application, procedure, and program version, indicating success when all three variables are matched.
- Detect IMAP buffer overflow attack.
- Detect FTP attempt login as root.

## 3 General Questions

#### 3.0.1 Any corrections or general comments about this lab?

#### 3.0.2 How long did it take you to complete this lab?

#### 3.0.3 Was it an appropriate length lab?